



Analog IC Design ASSESSMENT 5

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Submitted to
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1 Objective and Design Specifications

The objective is to design an inverting common-source amplifier with a PMOS current-source load using the look-up-table method. The NMOS driver and PMOS load are characterized separately, and the required widths are obtained from normalized device quantities rather than from a direct square-law design.

The design targets used in the supplied experiment material are summarized below.

Table 1: Design specifications for the current-source-load common-source amplifier.

Parameter	Specification
V_{DD}	1.8 V
C_L	1 pF
f_u	100 MHz
Minimum $ A_0 $	> 10 (20 dB)

The required transconductance is

$$g_m = 2\pi f_u C_L = 2\pi(100 \times 10^6)(1 \times 10^{-12}) \approx 6.283 \times 10^{-4} \text{ S} = 628.3 \mu\text{S}. \quad (1)$$

For the active-load stage,

$$|A_0| \approx \frac{g_m}{g_{ds,N} + g_{ds,P}}. \quad (2)$$

The look-up process therefore aims to select sufficiently high intrinsic gain while maintaining the required g_m and current density.

2 Look-Up Table Method

The normalized quantities used in the experiment are:

- g_m/g_{ds} for intrinsic-gain selection;
- g_m/I_D for selecting the operating point and obtaining the required drain current;
- I_D/W for converting current into physical width;
- g_{ds}/I_D for sizing the PMOS current-source load.

The main relations are

$$I_D = \frac{g_m}{g_m/I_D}, \quad W = \frac{I_D}{I_D/W}, \quad (3)$$

and for the PMOS load,

$$I_D = \frac{g_{ds}}{g_{ds}/I_D}, \quad W_P = \frac{I_D}{I_D/W}. \quad (4)$$

2.1 Intrinsic-gain selection

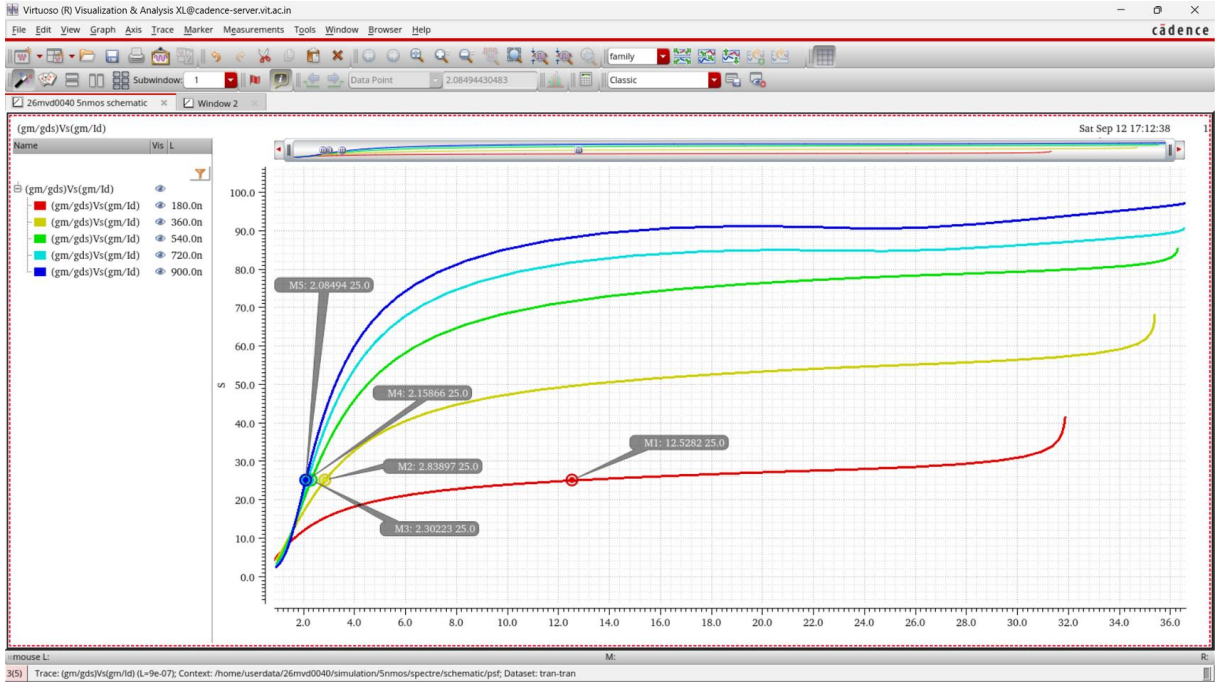


Figure 1: g_m/g_{ds} versus g_m/I_D for intrinsic-gain and channel-length selection.

2.2 NMOS current-density

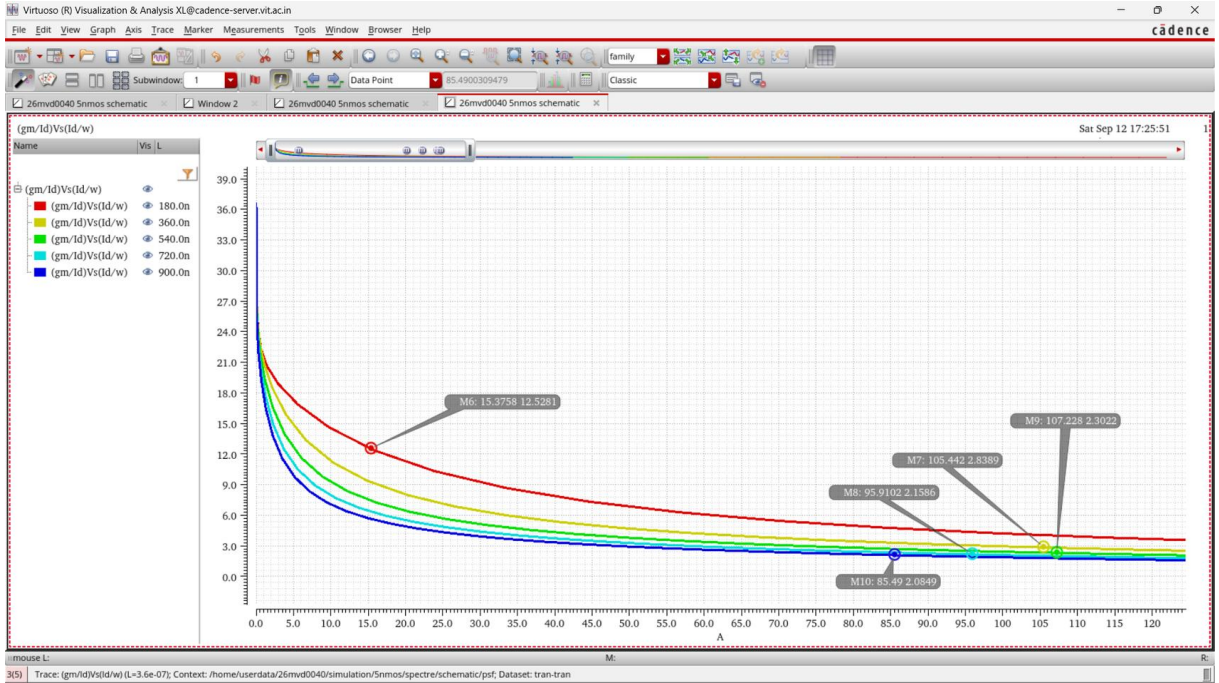


Figure 2: g_m/I_D versus I_D/W used for NMOS sizing.

2.3 PMOS output-conductance

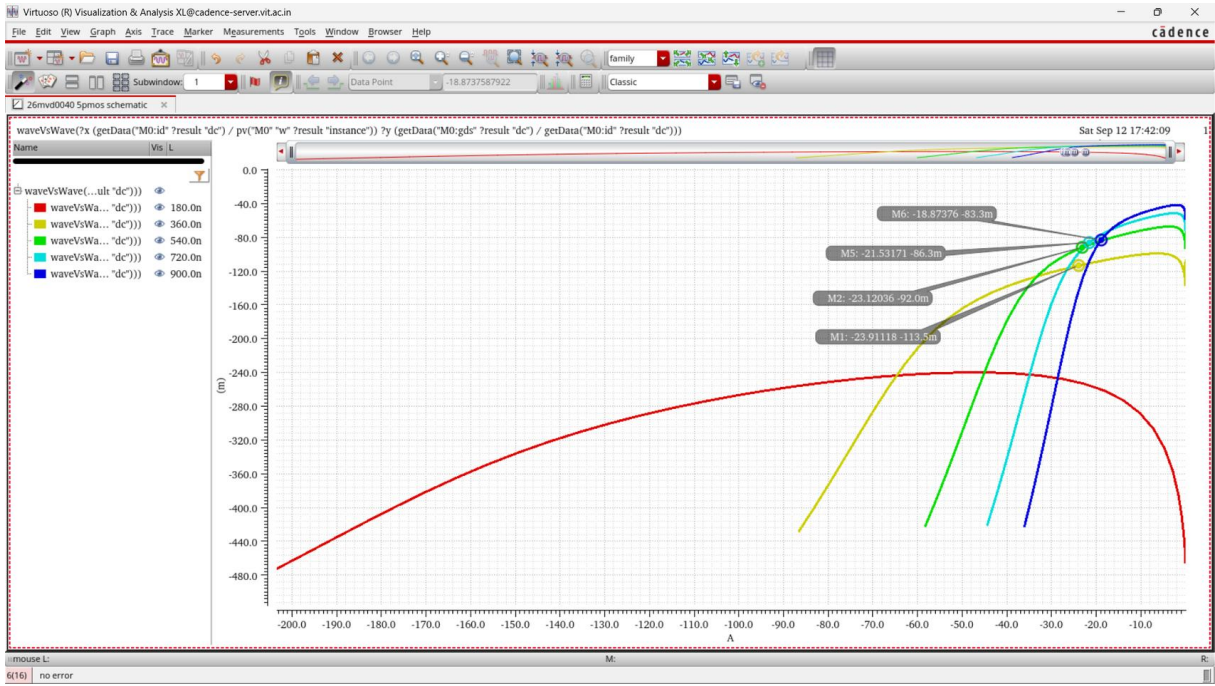


Figure 3: g_{ds}/I_D versus I_D/W used for PMOS sizing.

These three plots are included exactly once in the report and form the core of the look-up-table design flow.

3 NMOS Characterization and Width Selection

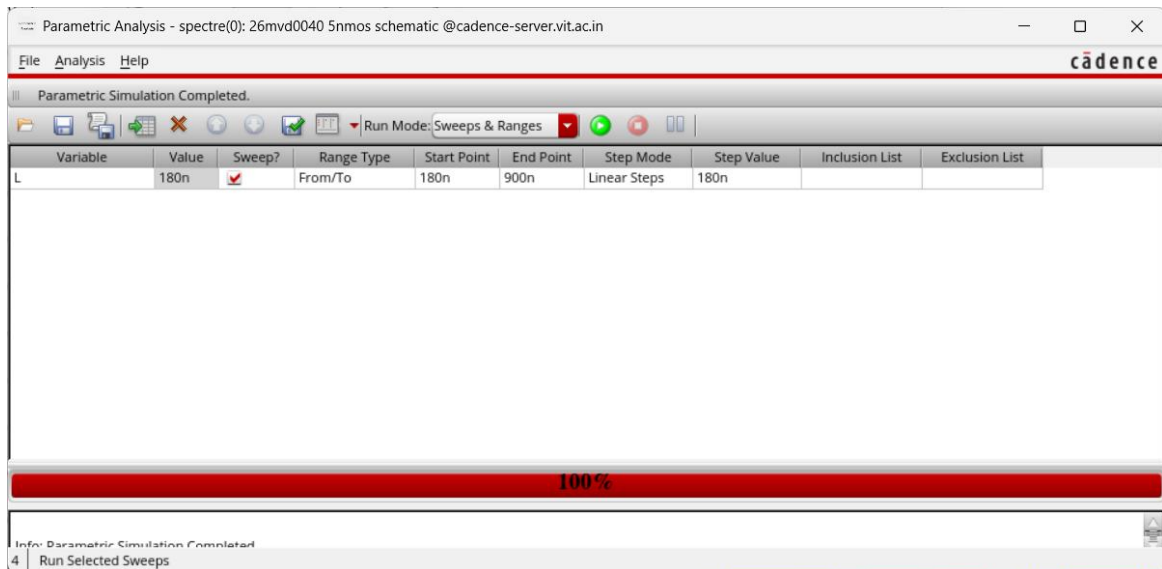


Figure 4: NMOS parameter setup from the supplied experiment material.

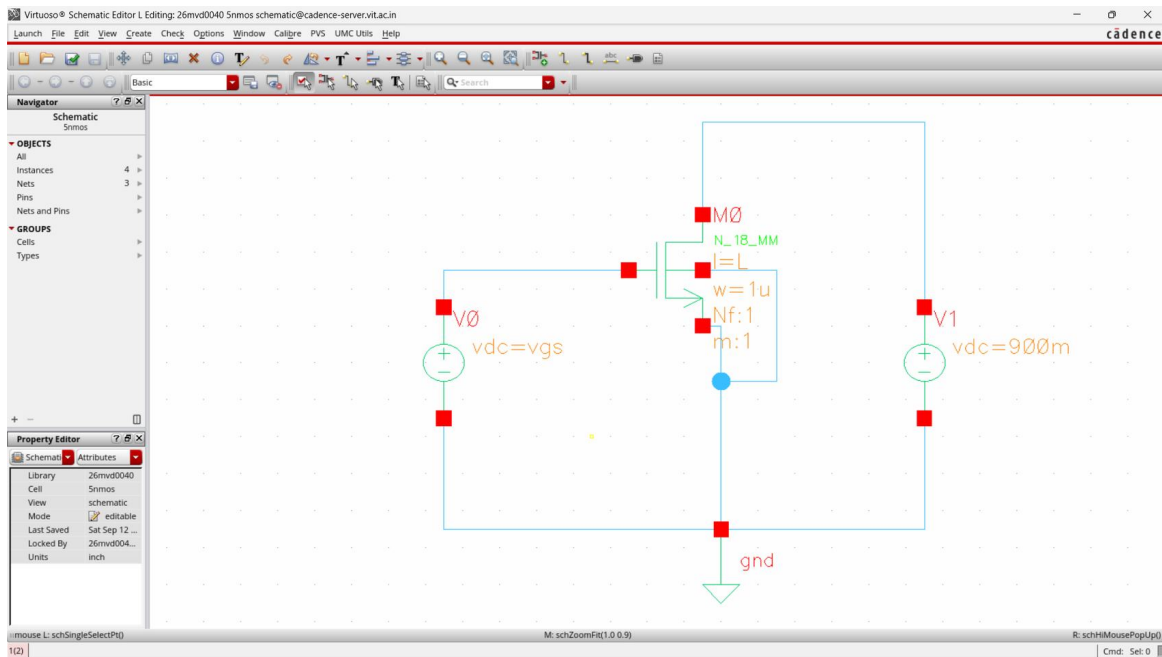


Figure 5: NMOS width-sweep characterization.

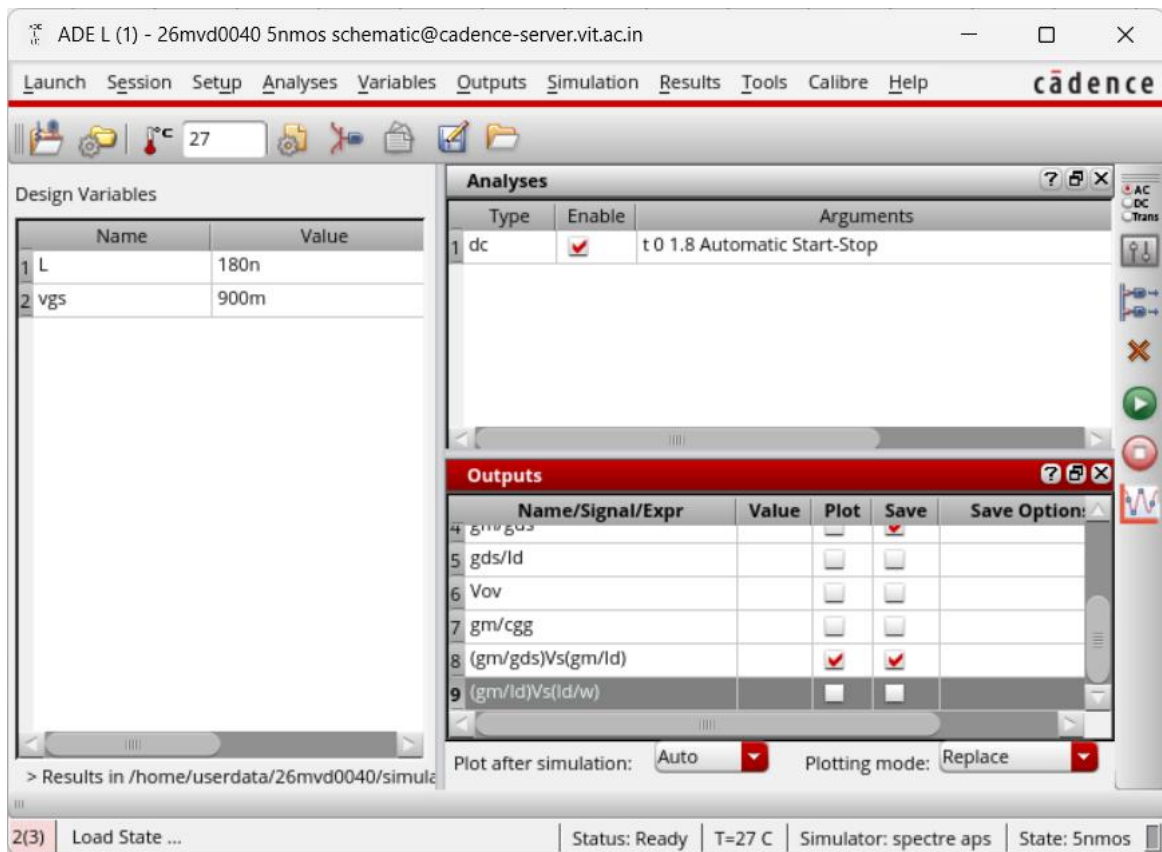


Figure 6: NMOS width-sweep analysis settings.

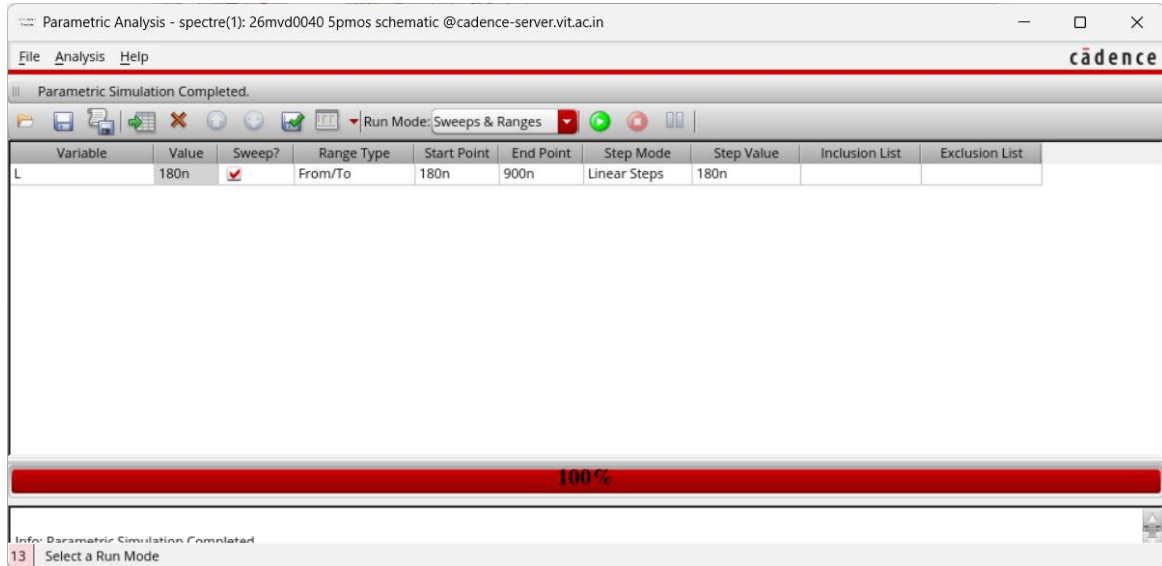
For the NMOS, the channel lengths considered are 180 nm, 360 nm, 540 nm, 720 nm, and 900 nm. The supplied design notes give the following current and width values.

Table 2: NMOS sizing values reported from the supplied look-up data.

L (nm)	I_D (μA)	W_N (μm)	Method	Use
180	50.12	3.2596	g_m/I_D , I_D/W	Characterization
360	221.21	2.0970	g_m/I_D , I_D/W	Characterization
540	272.78	2.5430	g_m/I_D , I_D/W	Characterization
720	290.92	3.0330	g_m/I_D , I_D/W	Characterization
900	301.213	3.5230	g_m/I_D , I_D/W	Characterization

The required physical width is determined from the current density at the selected operating point. The exact marker values visible in the supplied plot are retained in the source figures rather than recreated as a second graph.

4 PMOS Characterization and Width Selection

**Figure 7:** PMOS parameter setup from the supplied experiment material.

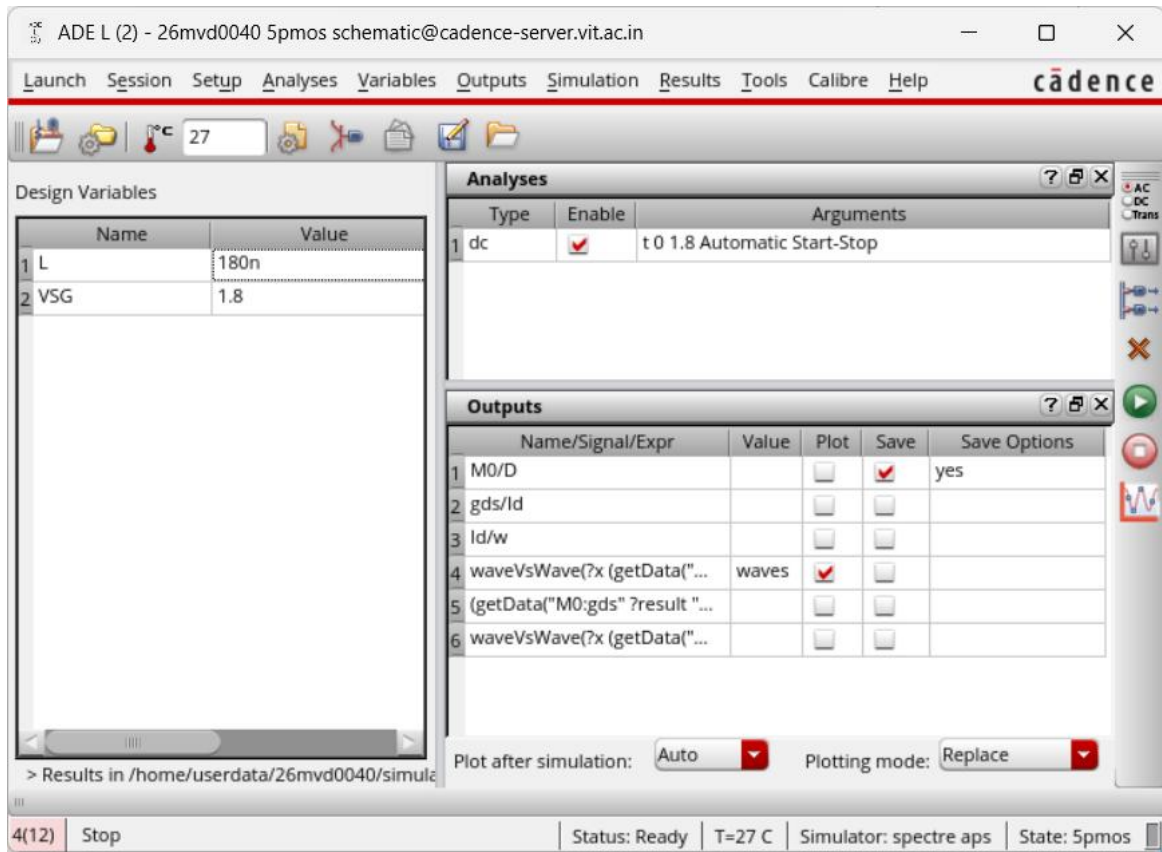


Figure 8: PMOS characterization/settings used for the look-up data.

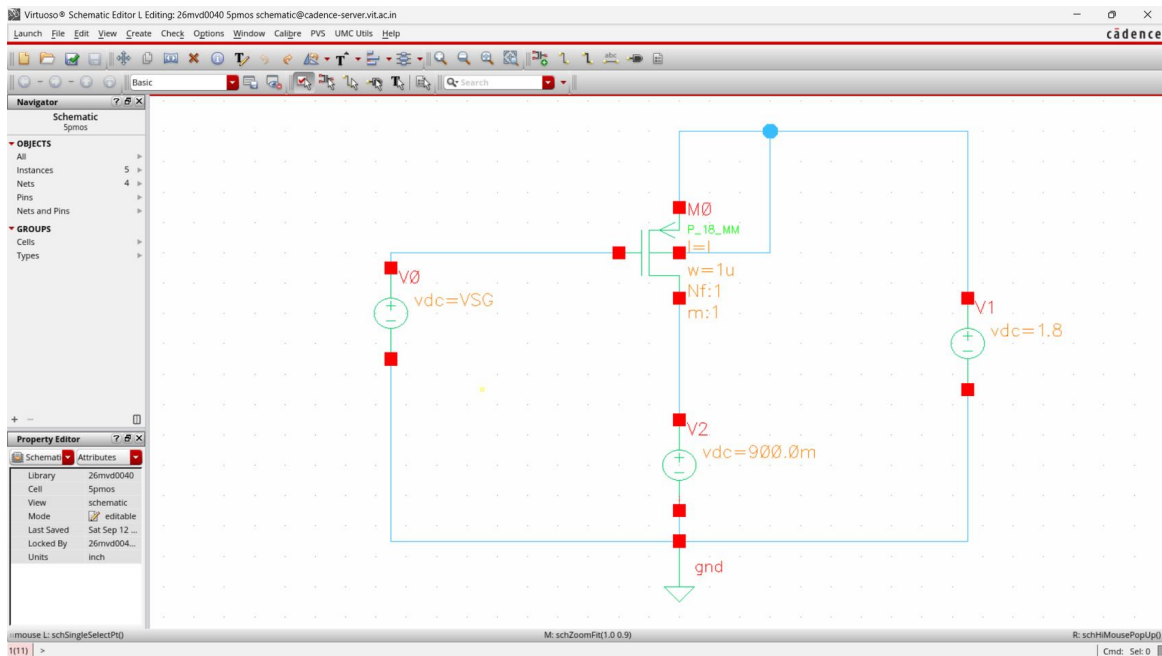


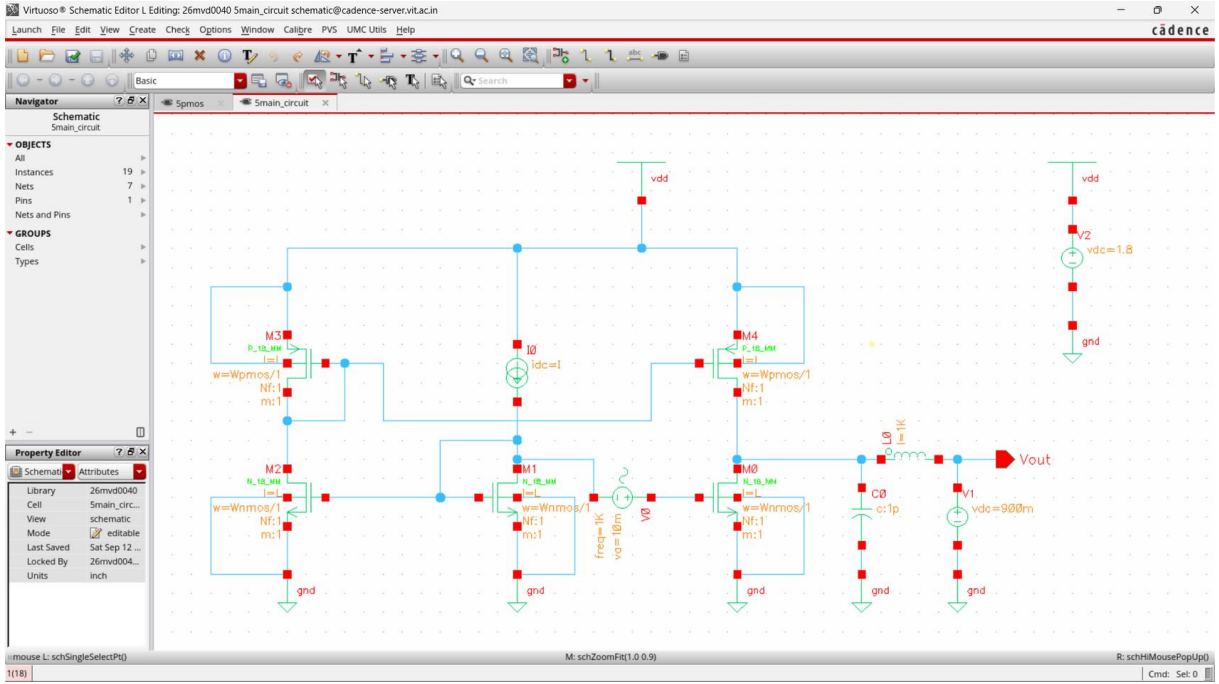
Figure 9: PMOS width-sweep characterization.

The supplied g_{ds}/I_D look-up data do not meet the selected target point at 180 nm. The usable PMOS values therefore begin at 360 nm. The reported values are:

Table 3: PMOS sizing values reported from the supplied look-up data.

L (nm)	g_{ds}/I_D (V^{-1})	I_D/W ($\mu A/\mu m$)	W_P (μm)	Status
180	—	—	—	Target point not reached
360	0.1135	23.911	9.25	Usable
540	0.0920	23.120	11.79	Usable
720	0.0863	21.531	13.511	Usable
900	0.0833	18.873	15.95	Usable

5 Final Current-Source-Load Circuit

**Figure 10:** Final common-source amplifier with PMOS current-source load.

The final circuit uses the NMOS common-source device as the signal transistor and the PMOS network as the active load. The output node is the common drain node. The sizing is performed separately for each selected channel length and then verified by the Cadence AC, transient, and pole-zero results.

6 Channel-Length Simulation Results

The supplied experiment archive contains four complete implementation sets: 360 nm, 540 nm, 720 nm, and 900 nm. For every length, the report includes exactly one A image, one B gain-in-dB plot, one C pole-zero output, and one settings screenshot.

6.1 360 nm

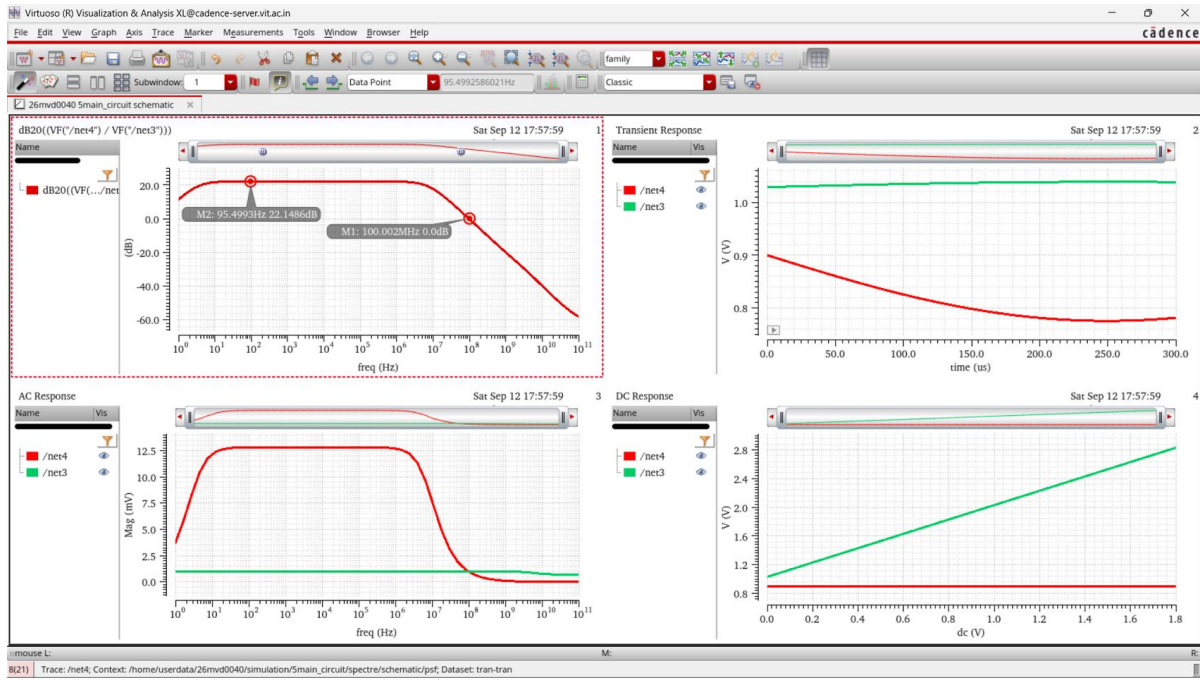


Figure 11: 360 nm simulation set – A.

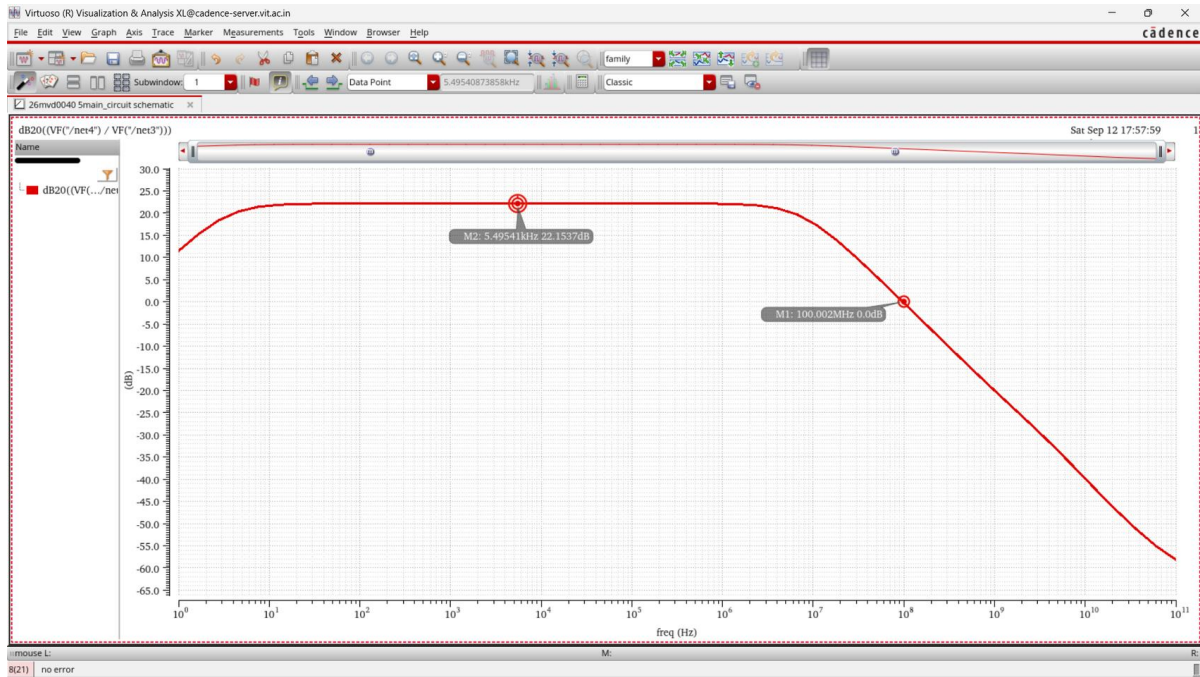


Figure 12: 360 nm gain response in dB – B.

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-3.27661e+00	0.00000e+00	5.00000e-01
2	-7.59967e+06	0.00000e+00	5.00000e-01
Zeros (Hz) at V(net4,0)/V0			
	Real	Imaginary	Qfactor
1	0.00000e+00	0.00000e+00	nan
2	-1.77098e+09	0.00000e+00	5.00000e-01
3	-9.65049e+09	0.00000e+00	5.00000e-01
4	1.19544e+11	0.00000e+00	-5.00000e-01
Constant factor = 1.20740e-24			
DC gain = 6.22445e-01			

Figure 13: 360 nm pole-zero result – C.

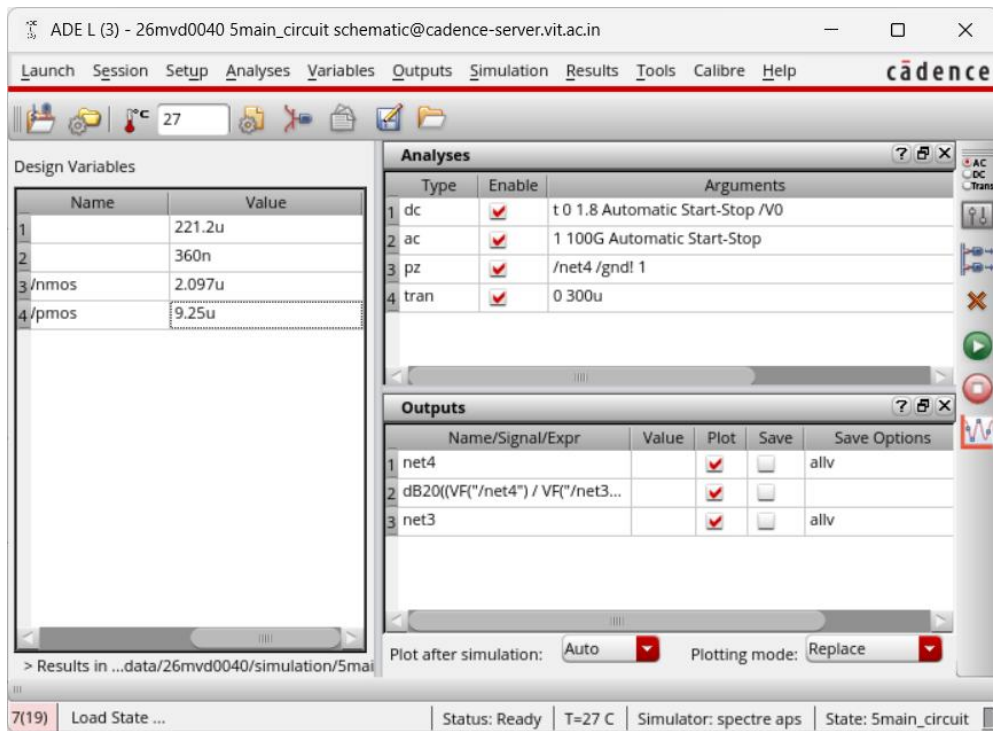


Figure 14: 360 nm Cadence simulation settings.

6.2 540 nm

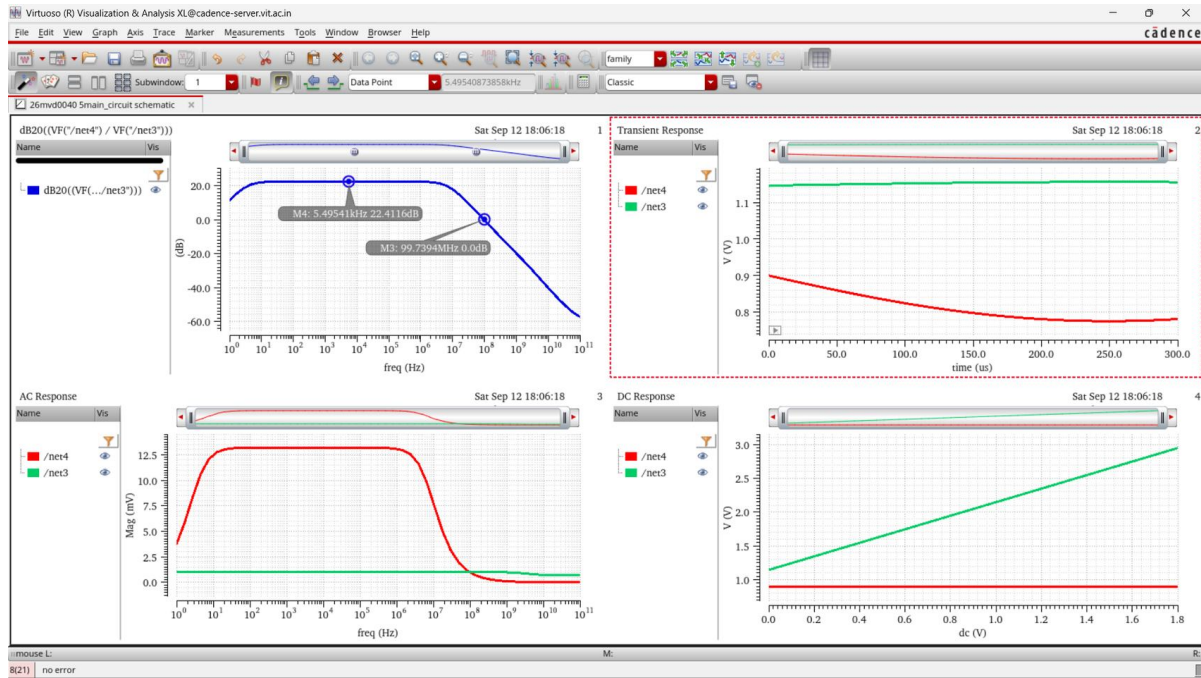


Figure 15: 540 nm simulation set – A.

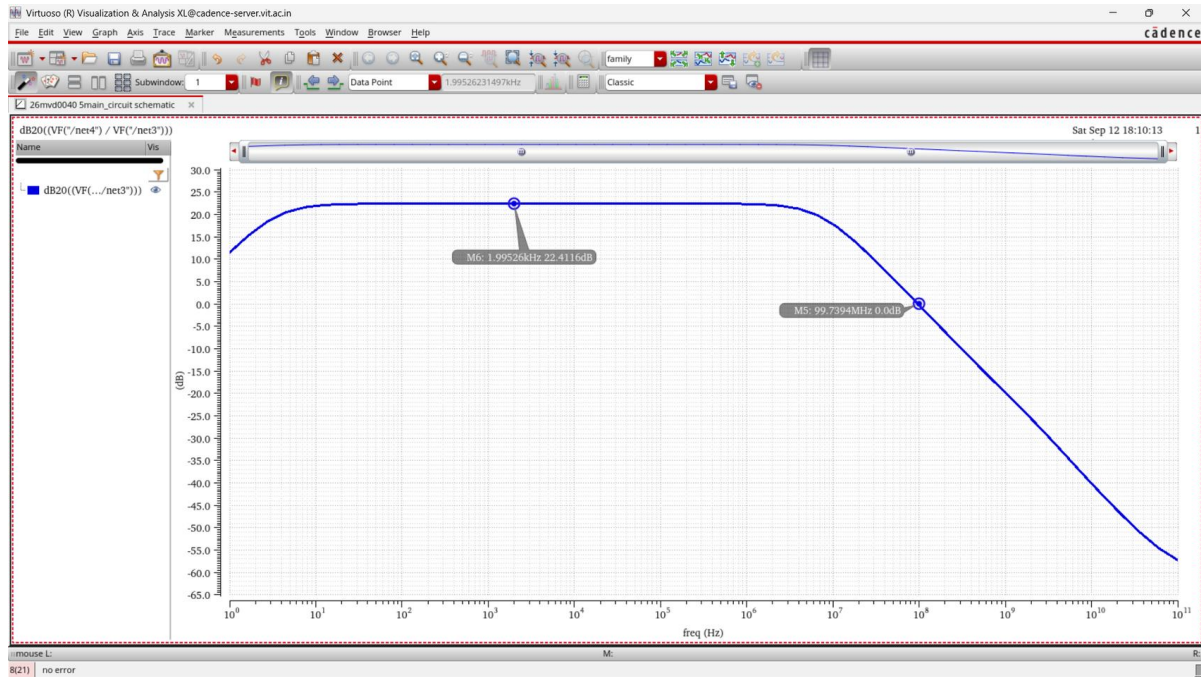


Figure 16: 540 nm gain response in dB – B.

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-3.37134e+00	0.00000e+00	5.00000e-01
2	-7.37592e+06	0.00000e+00	5.00000e-01

Zeros (Hz) at V(<u>net4</u> ,0)/V0			
	Real	Imaginary	Qfactor
1	0.00000e+00	0.00000e+00	nan
2	-1.03969e+09	0.00000e+00	5.00000e-01
3	-5.62973e+09	0.00000e+00	5.00000e-01
4	9.66793e+10	0.00000e+00	-5.00000e-01

Constant factor = 4.35827e-24

DC gain = 6.23164e-01

Figure 17: 540 nm pole-zero result – C.

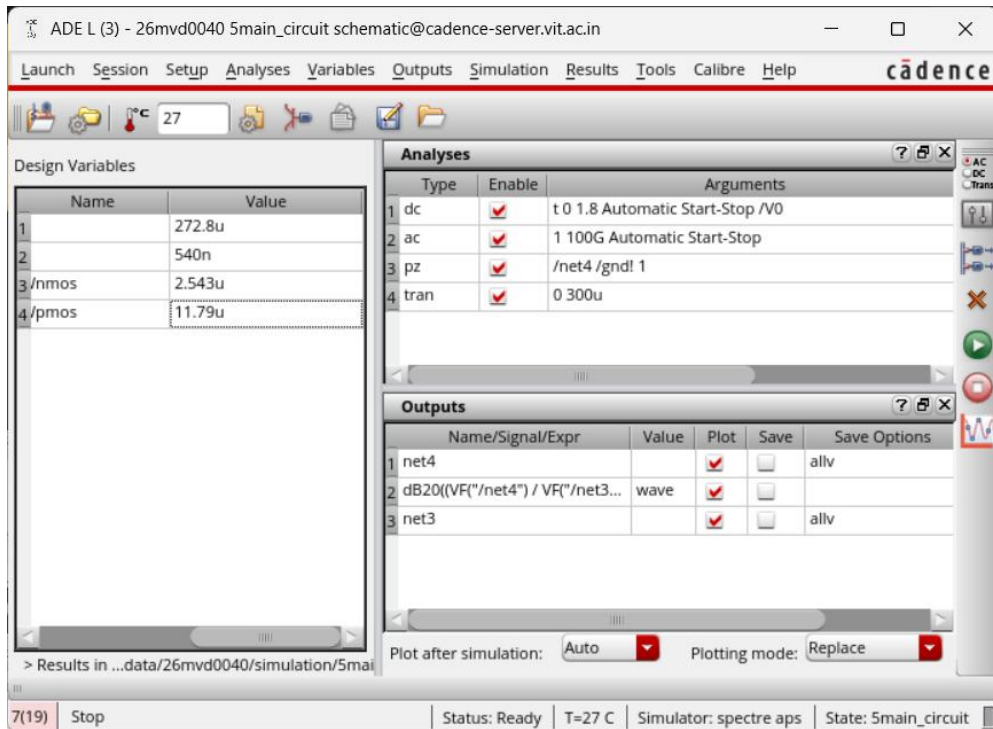


Figure 18: 540 nm Cadence simulation settings.

6.3 720 nm

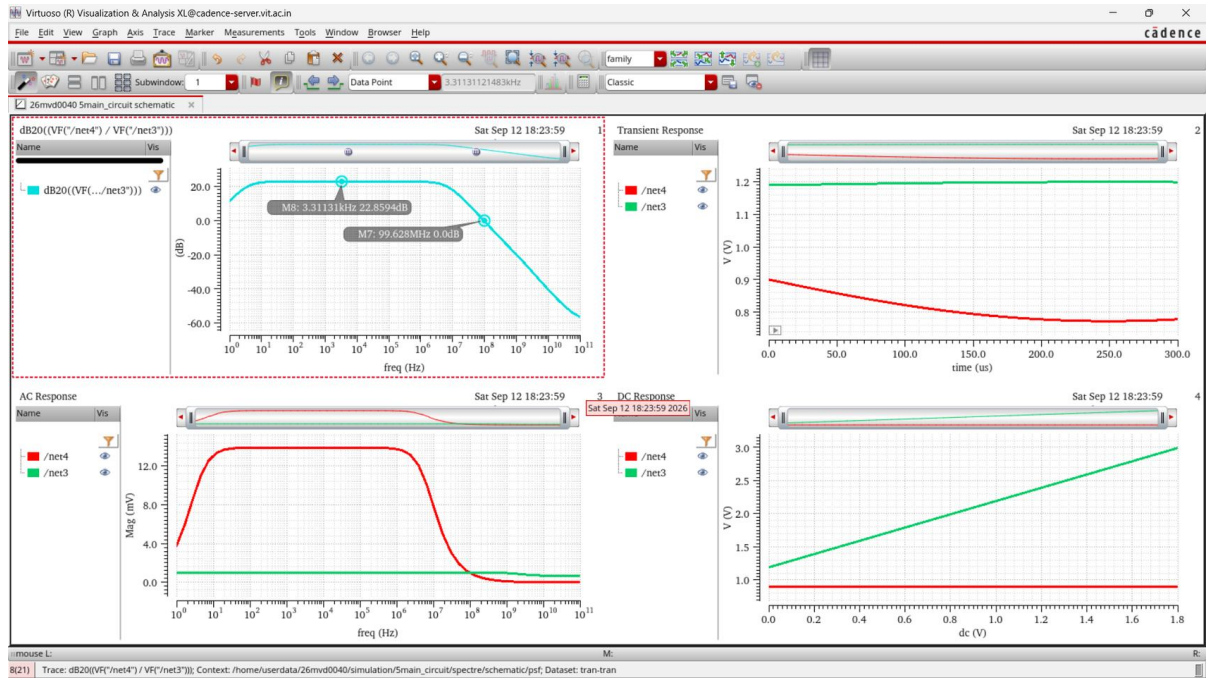


Figure 19: 720 nm simulation set – A.

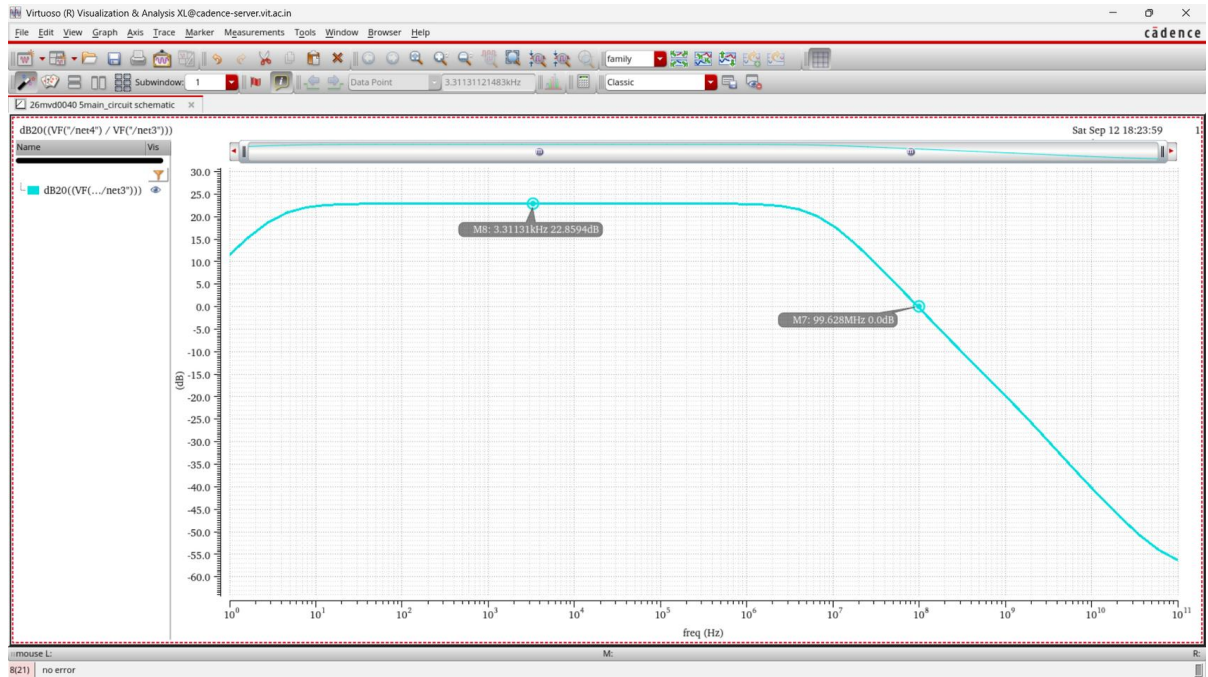


Figure 20: 720 nm gain response in dB – B.

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-3.54614e+00	0.00000e+00	5.00000e-01
2	-7.02223e+06	0.00000e+00	5.00000e-01
Zeros (Hz) at V(<u>net4</u> ,0)/V0			
	Real	Imaginary	Qfactor
1	0.00000e+00	0.00000e+00	nan
2	-7.35479e+08	0.00000e+00	5.00000e-01
3	-3.61369e+09	0.00000e+00	5.00000e-01
4	7.96088e+10	0.00000e+00	-5.00000e-01
Constant factor = 1.16844e-23			
DC gain = 6.23787e-01			

Figure 21: 720 nm pole-zero result – C.

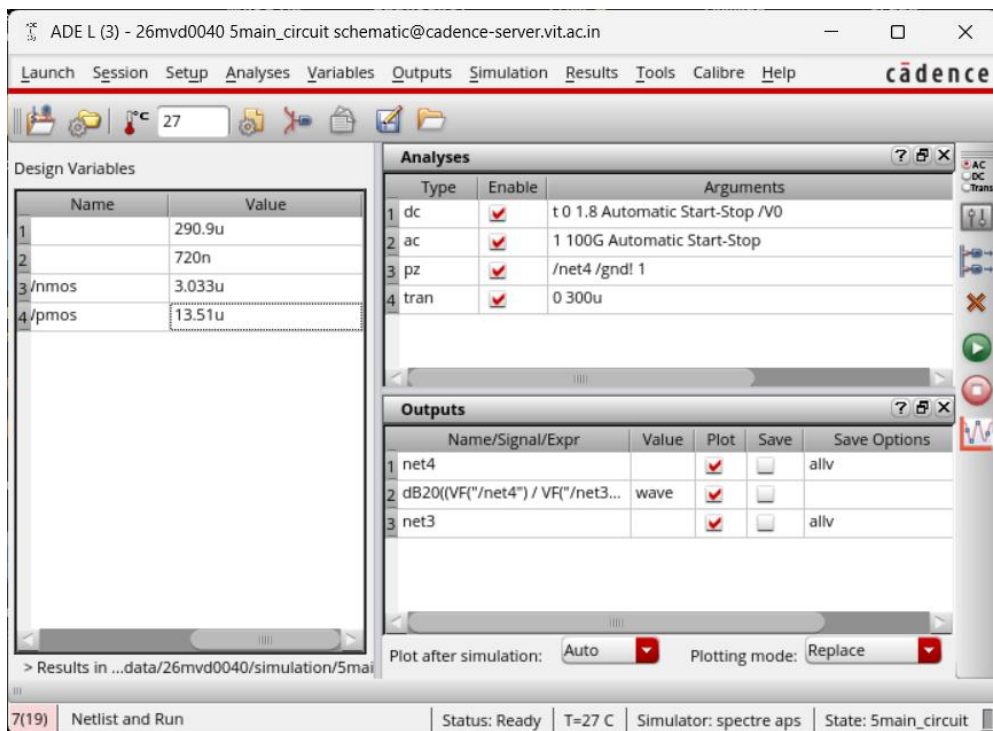


Figure 22: 720 nm Cadence simulation settings.

6.4 900 nm

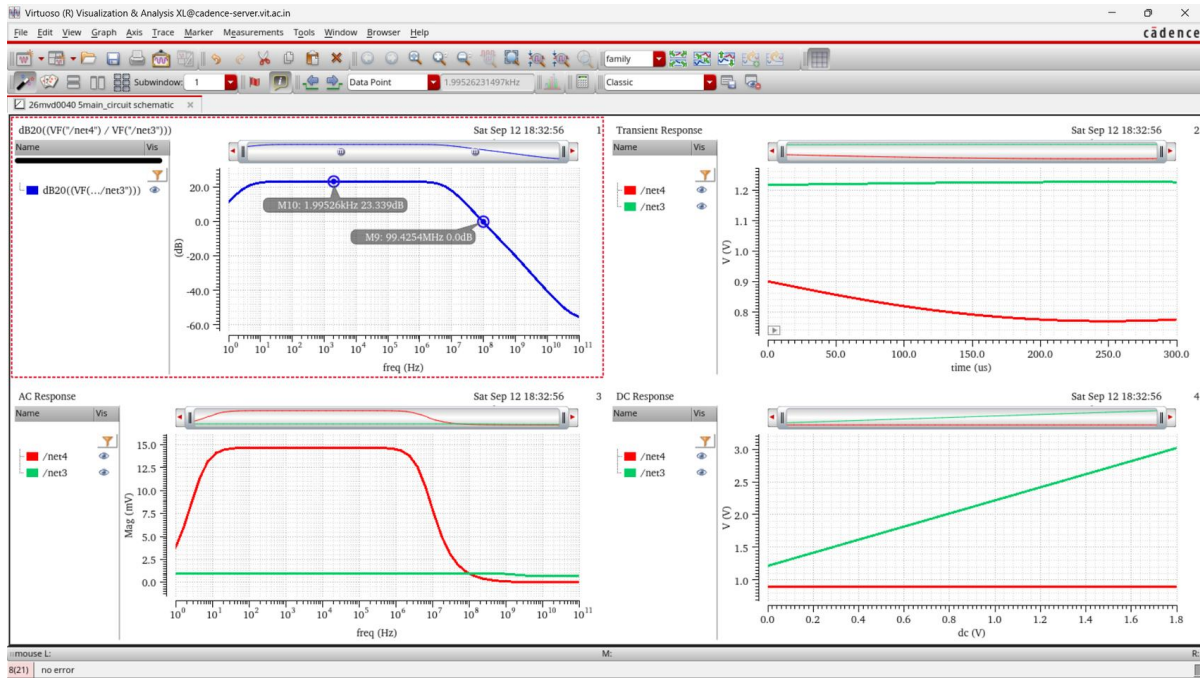


Figure 23: 900 nm simulation set – A.

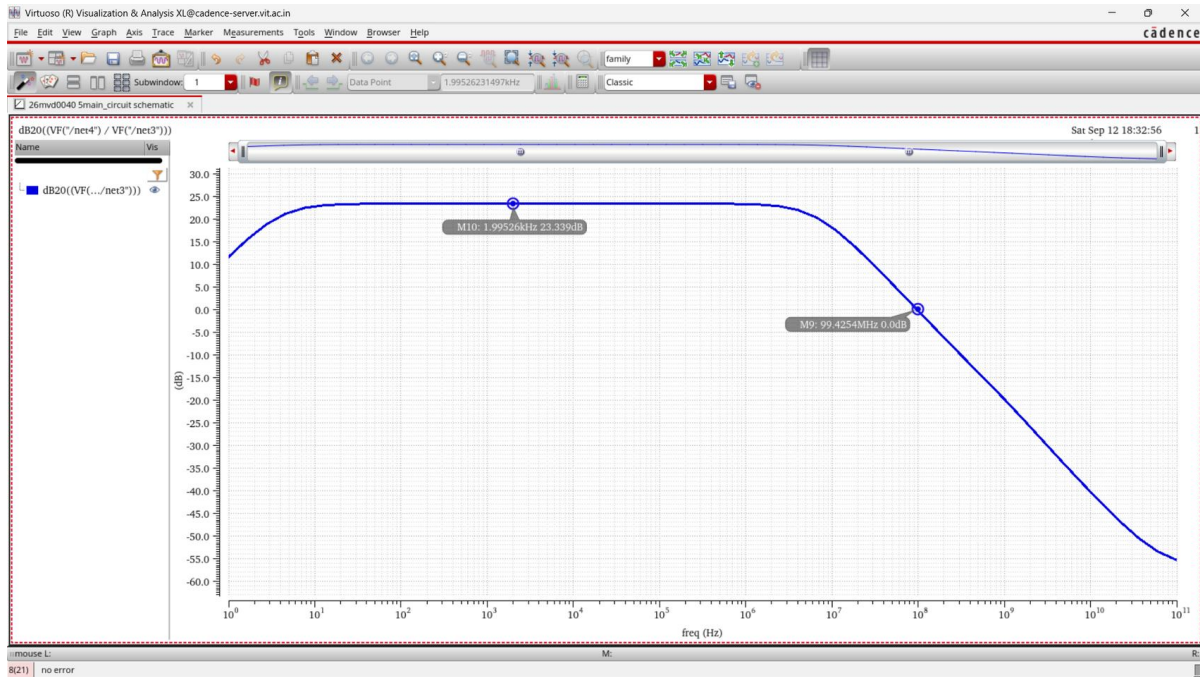


Figure 24: 900 nm gain response in dB – B.

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-3.74360e+00	0.00000e+00	5.00000e-01
2	-6.71564e+06	0.00000e+00	5.00000e-01
Zeros (Hz) at V(net4,0)/V0			
	Real	Imaginary	Qfactor
1	0.00000e+00	0.00000e+00	nan
2	-5.39790e+08	0.00000e+00	5.00000e-01
3	-2.50528e+09	0.00000e+00	5.00000e-01
4	6.70860e+10	0.00000e+00	-5.00000e-01
Constant factor = 2.75400e-23			
DC gain = 6.24425e-01			

Figure 25: 900 nm pole-zero result – C.

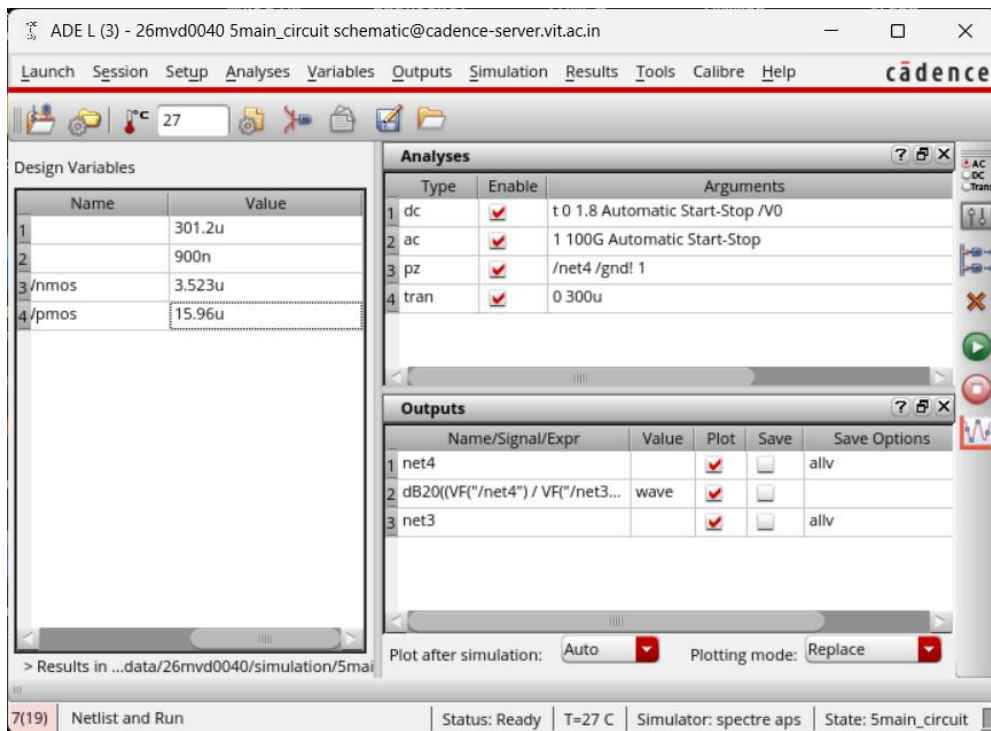


Figure 26: 900 nm Cadence simulation settings.

A Consolidated Look-Up Table Values

The table keeps the five channel lengths used in the final design flow and brings the NMOS and PMOS quantities into one place. The 180-nm PMOS row is retained to show the unsuccessful target-point selection, while the PMOS width is not used for that row.

Table 4: Single consolidated look-up table combining the design-point calculations.

L	g_m/g_{ds}	g_m/I_D	I_D/W N ($\mu\text{A}/\mu\text{m}$)	I_D N (μA)	W_N (μm)	g_{ds}/I_D P	I_D/W P ($\mu\text{A}/\mu\text{m}$)	W_P (μm)
180 nm	–	12.5281	15.3758	50.12	3.2596	0.5011	–	–
360 nm	–	2.8389	105.4423	221.21	2.0979	0.1135	23.911	9.25
540 nm	–	2.3022	107.2279	272.78	2.5439	0.0920	23.120	11.79
720 nm	–	2.1586	95.9101	290.92	3.0333	0.0863	21.531	13.51
900 nm	–	2.0849	85.4900	301.213	3.5234	0.0833	18.873	15.96

* Selected intrinsic-gain marker from the design notes: $g_m/g_{ds} = 25.2005$.

B Design Calculations

The handwritten calculations were converted into typed equations below. Only the calculation content is reproduced; the handwritten pages themselves are not inserted into the report.

B.1 Design target and required transconductance

The target values recorded in the design notes are

$$C_L = 1 \text{ pF}, \quad f_u = 100 \text{ MHz}. \quad (5)$$

Therefore,

$$g_m = 2\pi f_u C_L \quad (6)$$

$$= 2\pi(100 \times 10^6)(1 \times 10^{-12}) \quad (7)$$

$$\approx 628 \mu\text{S}. \quad (8)$$

For the current-source-load common-source amplifier,

$$|A_0| \approx \frac{g_m}{g_{ds,N} + g_{ds,P}}, \quad (9)$$

so the lookup procedure seeks a sufficiently large intrinsic gain while maintaining the required transconductance and acceptable current density.

B.2 NMOS drain-current calculation

For each selected length,

$$I_D = \frac{g_m}{g_m/I_D}. \quad (10)$$

Using $g_m \approx 628 \mu\text{S}$, the calculations are:

$$L = 180 \text{ nm} : I_D = \frac{628}{12.5281} = 50.12 \mu\text{A}, \quad (11)$$

$$L = 360 \text{ nm} : I_D = \frac{628}{2.8389} = 221.21 \mu\text{A}, \quad (12)$$

$$L = 540 \text{ nm} : I_D = \frac{628}{2.3022} = 272.78 \mu\text{A}, \quad (13)$$

$$L = 720 \text{ nm} : I_D = \frac{628}{2.1586} = 290.92 \mu\text{A}, \quad (14)$$

$$L = 900 \text{ nm} : I_D = \frac{628}{2.0849} = 301.213 \mu\text{A}. \quad (15)$$

B.3 NMOS width calculation

The physical width is obtained from

$$W_N = \frac{I_D}{I_D/W}. \quad (16)$$

Hence,

$$L = 180 \text{ nm} : W_N = \frac{50.12}{15.3758} = 3.2596 \mu\text{m}, \quad (17)$$

$$L = 360 \text{ nm} : W_N = \frac{221.21}{105.4423} = 2.0979 \mu\text{m}, \quad (18)$$

$$L = 540 \text{ nm} : W_N = \frac{272.78}{107.2279} = 2.5439 \mu\text{m}, \quad (19)$$

$$L = 720 \text{ nm} : W_N = \frac{290.92}{95.9101} = 3.0333 \mu\text{m}, \quad (20)$$

$$L = 900 \text{ nm} : W_N = \frac{301.213}{85.4900} = 3.5234 \mu\text{m}. \quad (21)$$

B.4 PMOS width calculation

For the PMOS active load, the current is matched to the selected NMOS operating point and the width is obtained from

$$W_P = \frac{I_D}{I_D/W}. \quad (22)$$

The 180-nm PMOS curve does not reach the selected g_{ds}/I_D target, so no final PMOS width is assigned at that length. For the usable lengths,

$$L = 360 \text{ nm} : W_P = \frac{221.21}{23.911} = 9.25 \mu\text{m}, \quad (23)$$

$$L = 540 \text{ nm} : W_P = \frac{272.78}{23.120} = 11.79 \mu\text{m}, \quad (24)$$

$$L = 720 \text{ nm} : W_P = \frac{290.92}{21.531} = 13.51 \mu\text{m}, \quad (25)$$

$$L = 900 \text{ nm} : W_P = \frac{301.213}{18.873} = 15.96 \mu\text{m}. \quad (26)$$

C Comparison and Conclusion

The look-up-table design uses g_m/g_{ds} to establish an adequate intrinsic-gain region, g_m/I_D to determine current for the required transconductance, and I_D/W to determine physical width. The PMOS current-source load is sized from the corresponding g_{ds}/I_D data.

The four final length-dependent simulation sets show the complete Cadence evidence available in the supplied archive. The B figures are the required gain-in-dB plots and are included once each for 360 nm, 540 nm, 720 nm, and 900 nm. The C figures provide the associated pole-zero outputs, while the settings figures document the corresponding analysis configuration.

The 180 nm PMOS case is excluded from final active-load sizing because the supplied g_{ds}/I_D curve does not reach the required target point. Among the usable channel lengths, increasing L improves output resistance and therefore tends to increase voltage gain, while the look-up based transconductance selection keeps the intended unity-gain-frequency region close to the target.